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Using attribute namespaces without repetition

Context

This paper provides wording for proposal P0028 as approved by Evolution Working Group (P0028R2 [1]). Please, note that paper P0028R2 provides rationale and change log.

Proposed Wording Changes

In 7.6.1/1, modify the production for *attribute-specifier* as follows:

```
...
attribute-specifier:
    [ [ attribute-using-prefixopt attribute-list ] ]
    alignment-specifier

attribute-using-prefix:
    using attribute-namespace :
...

```

Insert a new paragraph immediately after 7.6.1/1:

If an *attribute-specifier* contains an *attribute-using-prefix* the *attribute-list* following that *attribute-using-prefix* shall not contain an *attribute-scoped-token* and every *attribute-token* in that *attribute-list* is treated as if its *identifier* were prefixed with *N:*, where *N* is the *attribute-namespace* specified in the *attribute-using-prefix*. [Note: This rule imposes no constraints on how an *attribute-using-prefix* affects the tokens in an *attribute-argument-clause*.] [Example:

```
[[ using CC: opt(1), debug ]] void f() {}
    // Same as [[ CC::opt(1), CC::debug ]] void f() {}
[[ using CC: opt(1) ]][[ CC::debug ]] void g() {}
    // Okay (same effect as above).
[[ using CC: CC::opt(1) ]] void h() {}
    // error: cannot combine using and scoped attribute token

```

—end example]

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Jens Maurer identified issues in previous version of this paper.

References

[1] J. Daniel Garcia, Luis M. Sanchez, Massimo Torquati, Marco Danelutto, and Peter Sommerlad. Using non-standard attributes. Technical Proposal P0028R2, ISO/IEC JTC1/SC22/WG21, March 2016.